

December 2023

CY1006-1 – CHEMISTRY OF ENGINEERING MATERIALS

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Special properties of 'Li' metal that make it advantageous as an electrode material include**
A) Light weight
B) Low electrode potential
C) High electrical conductivity
D) All the above
- Which of the following is not the characteristic feature of a reserve battery**
A) Provides unlimited shelf life
B) Provides unlimited power supply
C) Highly reliable performance
D) Variety of design options
- Which of the following is INCORRECT statement for a fuel cell?**
A) Fuel cells need constant supply and removal of both reactants and the products.
B) They do not store chemical energy.
C) They are not rechargeable.
D) Fuel cells are rechargeable.
- In methanol-oxygen fuel cell, a membrane is inserted adjacent to the cathode on the inner side to**
A) Minimize the diffusion of methanol into the cathode
B) Minimize the diffusion of sulphuric acid into the cathode
C) Minimize the diffusion of methanol into the anode
D) Remove the produced CO_2
- Which one of the following is FALSE statement in a flow battery?**
A) Chemical energy stored in the form of active materials in electrolytes.
B) Electrolytes supplied at anode and cathode are separated by ion selective membranes.
C) Electrolytes are stored externally, generally in tanks.
D) Shedding of active materials stored at electrodes decrease its shelf life.
- Chemical formula of rust is**
A) $\text{Fe}_2\text{O}_4 \cdot x\text{H}_2\text{O}$
B) $\text{FeO} \cdot x\text{H}_2\text{O}$
C) Fe_3O_4
D) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- Electrolyte used for electroless copper plating is _____**
A) Copper
B) Cupric sulphate
C) Hydrogen sulphate
D) sodium chloride
- Caustic embrittlement is a type of.....**
A) Stress corrosion
B) Galvanic corrosion
C) Waterline corrosion
D) Pitting corrosion
- Potential of the calomel electrode depends on concentration of**
A) Hg
B) Hg_2Cl_2
C) KCl
D) KOH
- Glass electrode is an example for**
A) Metal-metal ion electrode
B) Reference electrode
C) Ion soluble electrode
D) Ion selective electrode

11. In concentration cells,
 A) Different electrodes are used
 B) Electrolyte concentrations are different
 C) Identical electrolyte concentration
 D) No electrodes are used
12. One of the important uses of polymethylmethacrylate is.....
 A) Cushions for furniture
 B) Lubricants
 C) Glazing automobiles and airplanes
 D) Light weight boat hulls
13. Epoxy resin prepared by the condensation of
 A) Epichlorohydrin and Bisphenol-A
 B) Epichlorohydrin and Diphenyl carbonate
 C) Bisphenol-A and Diphenyl carbonate
 D) Epichlorohydrin and Butane diol
14. Liquid crystals are usually composed of
 A) Coil like molecules
 B) Rod like molecules
 C) Oval molecules
 D) Triangular molecules
15. The array of tiny segments in LCD is
 A) Pixel
 B) Passive
 C) Active
 D) Count
16. By which properties, the orientation of molecules in a layer of liquid crystals can be changed?
 A) Magnetic field
 B) Electric field
 C) Electromagnetic field
 D) Gallois field
17. An example for two dimensional nanomaterials is
 A) Nanosheets
 B) Nanotubes
 C) Nanospheres
 D) Nanowires
18. Liquid crystal molecules for LCD applications must possess
 A) Single index of refraction
 B) No refractive index
 C) Two different indices of refraction
 D) None of the above
19. 1 nm =
 A) 10^{-7} m
 B) 10^{-9} m
 C) 10^9 m
 D) 10^{-3} m
20. Mesophase refers to:
 A) Solid state
 B) Liquid state
 C) Liquid crystal state
 D) Gaseous state

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is a reserve battery? Explain the construction and working of Na-ion battery	7	L2	1	1
	b) Explain the construction and working of PEM fuel cell.	5	L2	1	1
	c) List the advantages of PV cells.	4	L1	1	1
2.	a) Describe the differential aeration corrosion with suitable example.	8	L2	1	1
	b) Explain why anodized aluminum do not undergo corrosion?	2	L2	1	1
	c) What is anodic metal coating? Give the steps involved in Galvanization process.	6	L1	1	1
3.	a) Explain the construction and working of Vanadium redox flow battery.	6	L2	1	1
	b) Give any two differences between battery and fuel cells.	2	L1	1	1
	c) What is chemical conversion coating? Describe anodization of aluminum.	8	L2	1	1
Unit – II					
4.	a) What are concentration cells? Derive cell potential equation for concentration cells. Write the electrode reactions.	8	L1	2	1
	b) Explain the basic principle and working of optical sensors. What are their applications?	8	L2	2	1

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| 5. a) | Describe the principle and theory of potentiometric sensors using suitable example. | 8 | L2 | 2 | 1 |
| b) | Explain the synthesis properties and applications of the following polymers.
i) PMMA ii) Kevlar | 8 | L2 | 2 | 1 |
| 6. a) | Represent the cell formed by the coupling of two copper electrodes immersed in cupric sulphate solutions. The concentration of cupric ions in one electrode system is 100 times more concentrated than the other. Write the cell reaction and calculate the potential at 300K. | 6 | L3 | 2 | 1 |
| b) | Potential of calomel electrode depends on concentration of chloride ions. Justify. | 2 | L3 | 2 | 1 |
| c) | What are elastomers? Explain the synthesis and applications of Butyl rubber. | 8 | L1 | 2 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Describe the properties and synthesis of carbon nano tubes. | 8 | L2 | 3 | 1 |
| b) | Give the synthesis of quantum dot materials. | 8 | L1 | 3 | 1 |
| 8. a) | Explain the following properties of nanomaterials.
i) Surface area ii) Catalytic properties | 6 | L2 | 3 | 1 |
| b) | List any four effects of e-waste on environment and human health. | 4 | L1 | 3 | 1 |
| c) | Describe the methods of disposal and advantages of recycling e-wastes. | 6 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
